

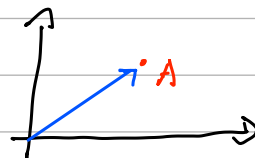
9/14 Ch. 1-2

Homogeneous System: All constants in linear equations are zero

Trivial Solution: A homogeneous of linear equations always have one solution $x_1=0$
 $x_2=0$, $x_3=0$

* A homogeneous system that has more variables than equations has many solutions, including trivial solution.

Vector space: 向量空間 R^n n -維度



Origin: 原點 O

position vector: 位置向量 - The vector from origin to a position.

initial point: 起點位置

terminal point: 終點位置

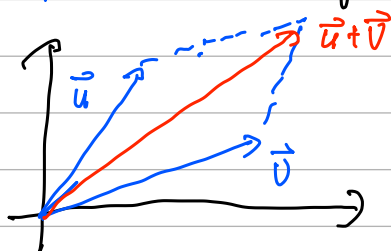
component: $(u_1, u_2, \dots, u_n)$ these sequences is call n -space or R^n

u_1 is first component - u_2 is second component

equal: If $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ two element of R^n and correspondingly same

Scalar multiplication: multiply a vector to a constant term

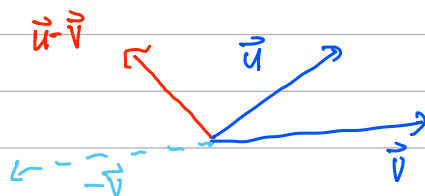
Addition:



Zero vector: $\vec{0}$. $\vec{u}$ has zero component

negative vector

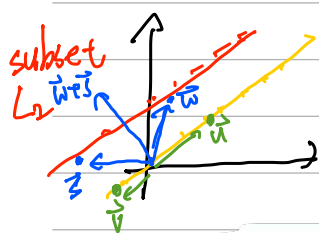
subtraction: $\vec{u} - \vec{v}$



Linear combination of vector: $\vec{u}_R = C_1 \vec{u}_1 + C_2 \vec{u}_2 + C_3 \vec{u}_3$, then $\vec{u}_R$ is the linear combination of $\vec{u}_1, \vec{u}_2, \vec{u}_3$ (線性組合)

subspace: 子空間: subset that is closed under addition and scalar multiplication.

subset 子集合 A group part of vector in the vector space.



L_1 subset

$L_1: \vec{u} + \vec{v}$ still in $L_1 \Rightarrow L_1$ is subspace of $\mathbb{R}^n$

$\vec{u} \times \vec{v}$ still in L_2

$L_2: \vec{w} \times \vec{v}$ still in $L_2 \Rightarrow L_2$ is subspace of $\mathbb{R}^n$

* All subspaces of $\mathbb{R}^n$ have the same features of vector space.

EXAMPLE 1 A subset of a vector space is often defined in terms of an arbitrary vector. Consider the subset W of $\mathbb{R}^3$ consisting of vectors of the form $(a, b, a + b)$, where the third component is the sum of the first two. The vector $(2, 5, 7)$ is in W for example, whereas $(2, 5, 9)$ is not. Let us show that W is a subspace of $\mathbb{R}^3$. Let $\mathbf{u} = (a, b, a + b)$ and $\mathbf{v} = (c, d, c + d)$ be vectors in W and k be a scalar. Then

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (a, b, a + b) + (c, d, c + d) \\ &= (a + c, b + d, a + b + c + d) \\ &= (a + c, b + d, (a + c) + (b + d))\end{aligned}$$

$\mathbf{u} + \mathbf{v}$ is in W since the third component is the sum of the first two components. W is closed under addition. Further,

$$\begin{aligned}k\mathbf{u} &= k(a, b, a + b) \\ &= (ka, kb, k(a + b)) \\ &= (ka, kb, ka + kb)\end{aligned}$$

$k\mathbf{u}$ is in W . W is also closed under scalar multiplication. W is a subspace of $\mathbb{R}^3$. It has all the algebraic properties of a vector space.

Let us look at the geometrical interpretation of this subspace W . Consider the arbitrary vector $\mathbf{u} = (a, b, a + b)$. Separate the variables a and b as follows:

$$\begin{aligned}\mathbf{u} &= (a, b, a + b) \\ &= (a, 0, a) + (0, b, b) \\ &= a(1, 0, 1) + b(0, 1, 1)\end{aligned}$$

The vectors $(1, 0, 1)$ and $(0, 1, 1)$ define a plane through the origin. See Figure 1.13. Any linear combination of vectors $\mathbf{u} = a(1, 0, 1) + b(0, 1, 1)$ lies in this plane. W is in fact the plane determined by the vectors $(1, 0, 1)$ and $(0, 1, 1)$.

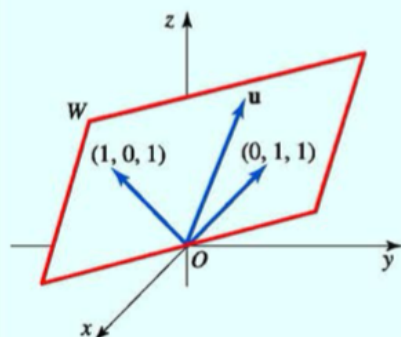


Figure 1.13 The plane determined by the vectors $(1, 0, 1)$ and $(0, 1, 1)$.